GOPAL KRUSHNA MAHAPATRA

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Work Experience

Frontend Development Intern Dec 2025 – Present

- Developed responsive and user-friendly web interfaces using HTML, CSS, JavaScript, and React, ensuring cross-browser compatibility.
- Implemented modern UI/UX practices, reusable components, and optimized layouts to enhance user experience and performance.
- Collaborated with designers and backend teams to integrate APIs and deliver scalable, production-ready frontend solutions.

Machine Learning Intern Dec 2025 – Present

- Developed and implemented machine learning models for data analysis and prediction, focusing on accuracy, scalability, and performance optimization.
- Performed data preprocessing, feature engineering, and model evaluation using Python and popular ML libraries to derive actionable insights.
- Collaborated on real-world problem statements, translating business requirements into data-driven solutions through analytical and statistical techniques.

Projects

IoT-Based Early Forest Fire Detection and Response System

- Designed and implemented an IoT-based forest fire detection system using ESP32, environmental sensors, and machine learning for early risk prediction.
- Integrated cloud platforms for real-time data monitoring, visualization, and automated alerts to enable rapid response.
- Improved fire detection accuracy and response time through multi-sensor data fusion and intelligent threshold-based decision making.

AI-Based Fruit Spoilage Detection System (Apple Case Study)

- Developed an IoT-enabled system to collect real-time environmental and gas sensor data from apples for spoilage analysis.
- Trained and evaluated machine learning models to predict spoilage stages based on sensor data patterns.
- Demonstrated early spoilage detection capability to support food quality monitoring and waste reduction.

AI-Storyteller

- Developed an AI-powered storytelling system using Gemini AI to generate coherent textual narratives and corresponding visuals from user prompts.
- Integrated automated image generation and story synthesis to create immersive, end-to-end storytelling experiences.
- Enabled multi-format content export, including PDF and audio, demonstrating expertise in AI integration and multimedia automation.

Education

2023 - Present	**Bachelor of Technology**	**NIST University**	**CGPA: 8 . 9 /10.0**
2021 - 2023	Class 12th - CBSE	K. C. Public School	**70%**
2020 - 2021	**Class 10th - CBSE**	**K. C. Public School**	**84%**

Publications

Mahapatra G. K., Padhi P., "*AI for Social Good & Resilience: Harnessing AI to Transform Society and Improve Quality of Life," Odisha AI Symposium 2025 Proceedings*, SOA University, 2025.

Skills

- Programming Languages: C, C++, Embedded C, Python, HTML5, CSS3, Java Assembly Language, Verilog
- Hardware & Tools: Microcontroller Boards, Sensors & Actuators, Integrated Circuits, Raspberry Pi
- Technologies: IoT, Digital Logic Design, Analog Circuit Design, Wireless Communication, 5G Technology, AI, ML, Cybersecurity
- Soft Skills: Leadership, Critical Thinking, Decision Making, Creativity, Problem Solving, Teamwork, Time Management
- Tools: Arduino IDE, Thonny IDE, Wokwi, VS Code, Colab, Jupyter, Github, Git Bash, Gemini, ChatGPT, Perplexity, Canva
- Platforms & Databases: Thingspeak, Blynk IoT, MQTT Broker, Vercel, Hugging Face, Tinkercad, Spabase, MongoDB Atlas, PostgreSQL